CURATED SOURCE DOCUMENT

Project Knowledge Base

Three portfolio project summaries prepared as a clear, structured source document for a Retrieval-Augmented Generation (RAG) test collection.

01	Real-Time Fraud Detection System
02	LungScope
03	StoryTime Machine

Document purpose: provide concise, attributable facts and terminology that can be retrieved accurately in question-answering experiments.

PROJECT 01

Real-Time Fraud Detection System

An anomaly-detection system for scoring simulated financial transactions in real time.

DATASET	PRECISION	API LATENCY
1,000+ simulated transactions	**92%**	**Under 200 ms**

PROJECT OVERVIEW

This project detects potentially fraudulent transactions using an Isolation Forest anomaly-detection model built with scikit-learn. It combines a FastAPI scoring service with a React dashboard and is packaged with Docker for consistent deployment.

KEY CAPABILITIES

- Applies Isolation Forest to identify anomalous transaction patterns in a dataset of more than 1,000 simulated transactions.
- Exposes transaction scoring through FastAPI endpoints with response times under 200 milliseconds.
- Reports 92% precision for fraud-detection performance.
- Provides a React dashboard for viewing the system's results.
- Uses Docker containerization for the application environment.

TECHNOLOGY AND IMPLEMENTATION

Machine learning: scikit-learn Isolation Forest. Backend/API: FastAPI. Frontend: React. Deployment packaging: Docker.

RAG RETRIEVAL NOTES

Useful query terms: fraud detection, anomaly detection, Isolation Forest, scikit-learn, simulated transactions, FastAPI, API latency, precision, React dashboard, Docker.

This page is a project-level source record intended for retrieval-augmented generation testing.

PROJECT 02

LungScope

An AI lung-disease diagnostic platform with image analysis and a full clinical workflow interface.

VALIDATION ACCURACY	AUGMENTATION	CORE MODEL
Approximately 94%	**+30% training data**	**Fine-tuned ViT**

PROJECT OVERVIEW

LungScope is an AI-powered platform for lung-disease diagnosis. Its FastAPI backend serves a fine-tuned Vision Transformer for image classification. The platform also uses DDPM diffusion models to augment training data and includes a React/Vite/MUI frontend for both diagnostic interaction and supporting workflow features.

KEY CAPABILITIES

- Serves a fine-tuned Vision Transformer through a FastAPI backend for lung-disease diagnostic inference.
- Uses DDPM diffusion models for data augmentation, increasing the training dataset by approximately 30%.
- Achieves approximately 94% validation accuracy.
- Supports image upload and Grad-CAM visualization for model interpretability.
- Includes authentication and an appointment workflow in the user-facing application.

TECHNOLOGY AND IMPLEMENTATION

Backend: FastAPI. Machine learning: fine-tuned Vision Transformer and DDPM diffusion models. Frontend: React, Vite, and Material UI (MUI). Features: upload, Grad-CAM visualization, authentication, and appointment workflow.

RAG RETRIEVAL NOTES

Useful query terms: LungScope, lung disease, diagnostic platform, Vision Transformer, ViT, DDPM, diffusion augmentation, FastAPI, Grad-CAM, React, Vite, MUI, validation accuracy.

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PROJECT 03

StoryTime Machine

A narrative version-control engine for writers of serialized audio shows, built during the Pocket FM Zero to One Hackathon.

BUILD TIME	TEAM	HACKATHON RESULT
36 hours	**3 people**	**Top 50 / 1,000+ teams**

PROJECT OVERVIEW

StoryTime Machine is a narrative version-control engine designed for writers creating serialized audio shows. The project was built by a three-person team in 36 hours for Pocket FM's Zero to One Hackathon, with Codex used during development. It was recognized among the Top 50 of more than 1,000 teams nationwide.

KEY CAPABILITIES

- Addresses the needs of serialized audio-show writers through a narrative version-control concept.
- Was developed in a 36-hour hackathon sprint by a three-person team.
- Used Codex as part of the build process.
- Competed in the Pocket FM Zero to One Hackathon.
- Placed in the Top 50 among more than 1,000 teams nationwide.

TECHNOLOGY AND IMPLEMENTATION

Product focus: narrative version control for serialized audio storytelling. Event: Pocket FM Zero to One Hackathon. Development context: 36-hour sprint, three-person team, Codex-assisted development.

RAG RETRIEVAL NOTES

Useful query terms: StoryTime Machine, Pocket FM, Zero to One Hackathon, narrative version control, serialized audio shows, writers, Codex, 36 hours, three-person team, Top 50.

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